Electric Motor



Category: **Electrical & Electronic Components & Kits** Type: **Full SOP** Priority: **High**

8 labs

Covers these lab items: AC and DC motors; BLDC Motors; MOTER ELECTRIC 1 PHASE; MOTER ELECTRIC 1 PHASE ISI Mark; Motor

What it is used for

An electric motor converts electrical energy into mechanical rotation or motion. Correct electrical connections, suitable load, and proper cooling help prevent electric shock, overheating, and mechanical damage.

Main parts

- Stator provides the stationary magnetic field or winding assembly.
- Rotor rotates inside the motor to produce mechanical output.
- Shaft transfers rotation to the driven equipment.
- Bearings support the shaft and reduce friction.
- Motor housing protects internal components and supports heat dissipation.
- Terminal box or connectors provide electrical connection points.
- Cooling fan or vents remove heat on applicable models.

Before you start

1. Read the motor nameplate and manual to confirm voltage, phase, speed, and duty rating.
2. Allow only trained and authorised personnel to make electrical connections.
3. Isolate the power supply and verify it is safe before inspection or wiring.
4. Inspect the housing, shaft, mounting bolts, terminal box, cables, and cooling vents.
5. Check that the motor is firmly mounted and the driven equipment is correctly aligned.
6. Confirm guards are fitted over couplings, belts, fans, and other moving parts.

How to use

1. Confirm the motor and connected equipment are suitable for the intended task.
2. Check the nameplate and approved wiring diagram before connecting supply terminals.
3. Ensure earthing, protective devices, and cable ratings are appropriate for the installation.
4. Secure all permitted electrical connections and close the terminal box before energising.
5. Clear tools and loose objects from the motor and surrounding area.
6. Switch on the supply using the approved control and observe the starting behaviour.
7. Check rotation direction, vibration, noise, and temperature within the manufacturer's limits.
8. Stop the motor immediately if it overheats, vibrates excessively, or produces unusual noise or smell.
9. Isolate power and wait for moving parts to stop before adjustment or maintenance.
10. After use, switch off the supply and record any faults for authorised personnel.

⚠ Safety

- Never touch live terminals or internal motor parts while the supply is connected.
- Do not bypass earthing, guards, overload protection, or other safety devices.
- Keep hands, hair, clothing, and tools away from rotating shafts and couplings.
- Never connect a motor to a supply that does not match its nameplate requirements.
- Allow hot motors to cool before touching the housing or nearby components.
- Use appropriate lockout and isolation procedures before maintenance.

Care & storage

- Keep cooling vents and the motor exterior free from dust and obstruction.
- Inspect mounting bolts, couplings, cables, and guards regularly for wear or looseness.
- Follow the manual for lubrication, bearing inspection, and preventive maintenance.
- Arrange authorised inspection for overheating, insulation damage, unusual vibration, or repeated tripping.

Common mistakes to avoid

- Connecting the motor without checking its nameplate and wiring diagram.
- Running the motor with missing guards or loose mounting bolts.
- Ignoring incorrect rotation, excessive vibration, overheating, or unusual smells.
- Opening the terminal box before isolating and verifying the power supply is safe.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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